

# Omid Ardalani

Computational Biologist | Genomes at Scale, Machine Learning on Genomes, Agentic AI Systems for Biology

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## PROFILE

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Ten years ago I was the first person in my department to do computational biology. There was no course to take and nobody to learn it from, so I taught it to myself, because I believed a new Darwin would come out of computational biology: computation was the only way to look at life at a scale where the real patterns become visible, and nothing else could give us that perception. A decade later I have built the largest strain-resolved collections of in-silico bacterial cells assembled anywhere, and the pipelines that put that scale within reach of other people. My conviction has since changed shape. The data is here. What biology lacks is a reasoning system that can work at this scale, and that is what machine learning, and now language models, change. My work on non-negative matrix factorization of bacterial genomes recovers signals nobody imagined were recoverable, and the agentic system I built on Claude Code mines public databases at a resolution nobody had the throughput to reach. The knowledge was already sitting there.

## SELECTED ACHIEVEMENTS

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- **Scale.** Built the largest strain-resolved genome-scale metabolic model collections assembled for a bacterial species and for a bacterial family: roughly 12,000 *E. coli* and 2,300 Lactobacillaceae in-silico twins, and the open pipelines that make that scale reachable.
- **A verifiable signal in biology.** Every one of those models emits falsifiable predictions: this strain grows on this medium or it does not, this gene is essential or it is not. Each is scoreable against experimental measurement, which makes it one of the few parts of biology with a clean problem-and-answer pair, and therefore usable as a training and evaluation signal.
- **A tool biologists actually use.** First author on the pangenome-scale model-generation pipeline behind **BiGG 2026** (biggr.org), the next release of the BiGG Models database, the reference model repository for the field.
- **Agentic AI for biology.** Designed and operate a multi-agent research system on Claude Code that does real life-science work (bioinformatics pipelines, database mining, figure generation, literature synthesis), with grounded verifiers that check each claim against the underlying data instead of asking the model whether it was right.
- **The genetic basis of metabolic function.** First author, *Science Advances* (2026). Now scaled into a pipeline that searches millions of genes across thousands of genomes, tests whether each gene is genuinely functional rather than merely present, and maps the full space of distinct genetic codes that encode one and the same biochemical function.
- **From bench to product.** Isolated seven lactic acid bacteria strains and led their scale-up to commercial production at a biotech startup. Lactobacillaceae-related patent. Main inventor; Notice of Invention filed, DTU. Trained in the group of **Prof. Bernhard O. Palsson**.

## KEY SKILLS

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- **Machine learning for biology:** protein language models and embeddings; ML prediction of enzyme kinetics (kcat, Km); unsupervised learning and non-negative matrix factorization (NMF) of pangenomes; PyTorch, TensorFlow, scikit-learn; model training, validation, and benchmarking against reference data
- **Agentic AI systems and LLM tooling:** heavy Claude Code user; designed and run a multi-agent research system for biology (tool scaffolding, subagent orchestration, MCP servers, hooks, reusable skills); evaluation harnesses and grounded verifiers; eval-driven development; context and prompt engineering
- **Software engineering and infrastructure:** production Python (advanced), R (advanced), Bash, Linux; large codebases; open-source release and maintenance (github.com/omidard); Git; Snakemake and Nextflow; HPC (SLURM) and cloud (Azure); large-scale distributed batch computing; APIs, databases, and public web deployment
- **Genomics and bioinformatics:** pangenome reconstruction across whole bacterial families; sequence search and clustering (BLAST, DIAMOND, HMMER, MMseqs2, CD-HIT); annotation (Bakta, Prokka, eggNOG, InterProScan, Pfam); gene-family and orthology analysis; taxonomy and ANI (GTDB-Tk); horizontal gene transfer; AMRFinderPlus; Biopython; protein structure at scale (Foldseek, US-align, SWISS-MODEL)

- **Biological databases and their failure modes:** UniProt and Swiss-Prot, NCBI, GTDB, KEGG, BiGG, MetaNetX, AlphaFold DB; schema-level work, and a documented record of finding the silent errors these resources propagate (duplicate assemblies, contradictory taxonomy, homology-derived annotations that are chemically impossible)
- **Metabolic modeling:** genome-scale and pangenome models (GEM, panGEM); FBA and FVA; COBRApy, CarveMe, MEMOTE; computational strain design (OptKnock, FSEOF); in-silico media formulation
- **Laboratory and translation:** microbiology, molecular biology, strain isolation and development, fermentation and bioprocess, QA/QC, GMP/GLP, technology transfer; communicating computational results to experimental scientists and to business

## PROFESSIONAL EXPERIENCE

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### Postdoctoral Researcher, DTU Biosustain / DTU BRIGHT (Technical University of Denmark)

Kgs. Lyngby, DK · 2025-Present

- Lead the pangenome-scale program that searches **millions of genes across thousands of bacterial genomes**, tests whether each gene is genuinely functional rather than merely annotated, and maps the complete space of genetic solutions encoding the same metabolic function.
- First author on the pipeline behind **BiGG 2026, the next release of the BiGG Models database** (biggr.org), the field's reference repository for genome-scale models: open, versioned, and reproducible.
- Own the production infrastructure: Python pipelines and large codebases, Nextflow and Snakemake workflows, HPC and Azure batch compute, open-source releases, and public web browsers over the model collections.
- Built the **verification layer the group now reuses**, after establishing that standard annotation practice produces systematic, silent errors: duplicate assemblies inflating genome counts (16,404 records reduced to 8,826 truly unique), taxonomy that disagrees between NCBI and GTDB, gene-symbol-to-function calls that are artifacts, and homology-inherited gene-to-reaction assignments that are chemically impossible. Each was found by grounding the claim in data rather than trusting the database.
- Apply **protein language-model embeddings and ML kcat/Km predictors** to prioritize enzymes, and **non-negative matrix factorization** to recover genome structure that no other method surfaced.

### Independent Work, Agentic AI Systems for Biology

2025-Present

- Designed and operate a **multi-agent research system built on Claude Code** that carries out real computational biology: bioinformatics pipelines, public-database mining, analysis notebooks, figure generation, and literature synthesis, organized as specialized agents with explicit contracts, handoffs, and persistent project state.
- Built the **evaluation layer** that makes its output trustworthy: grounded verifiers that measure the artifact itself (reading a rendered figure back and measuring font sizes, overlaps, clipping, and blank panels, rather than asking the model whether the figure is good), reviewer agents held separate from the author agent and given isolated context, and checks requiring every claim to resolve to source data.
- **Core principle: a model does not know when it is wrong, so the system must verify against the world, not against the model.** Every subsystem carries an eval harness whose pass or fail criterion is computed from data.
- Engineering delivered: tool scaffolding and integrations, MCP servers, hook-driven routing, subagent orchestration, reusable skills, and reproducible workflow graphs.

### Researcher, Computational Biology, DTU

2024-2025

- Developed marker-gene and rare-gene methods, supervised and unsupervised, that classify strains and identify niche-adaptive metabolism across pangenomes.

### Ph.D. Researcher, Systems Biology, DTU (group of Prof. Bernhard O. Palsson)

2021-2024

- Built the largest strain-resolved metabolic model collections assembled for a bacterial species and family, roughly **12,000 *E. coli*** and **2,300 Lactobacillaceae genome-scale models**, and wrote the reconstruction pipelines that made that scale possible. Released as open-source software (github.com/omidard/EcopanGEM) and as public model browsers (omidard.github.io/panGEMs).
- Predicted strain-specific growth and metabolite production, including short-chain fatty acids, and formulated growth media in silico.

- As **first and corresponding author** on a multi-institution study, set the research direction, coordinated the co-authors, and carried the work through peer review. **Led the Lactobacillaceae pangenome study (mSystems)** as lead scientist.

## **R&D Specialist, Strain Development, Fermentation and QA/QC, Behdad Genetics Corp. (biotech startup)**

Tehran, IR · 2019-2021

- **Isolated seven lactic acid bacteria strains and led their scale-up from laboratory to commercial production** as feed probiotics, running a full design-build-test-learn and technology-transfer cycle.
- Designed and optimized growth media and fermentation processes; established QA/QC and authored SOPs to GMP and GLP standards. Hands-on molecular biology, microbiology, and bioprocess work at the bench and at production scale, which is where I learned what a real scientific workflow looks like and where it breaks.

## **SELECTED OPEN-SOURCE TOOLS**

- **BiGG 2026** (biggr.org): the next-generation BiGG Models database. **panGEMs** (omidard.github.io/panGEMs): public browser over 2,313 *E. coli* and 2,346 Lactobacillaceae genome-scale models. **EcopanGEM** and **LactoPanGEM**: pangenome-scale reconstruction of *E. coli* and Lactobacillaceae metabolism. **PanGPSR**: gene to protein to structure to reaction at pangenome scale. **GrowthDB**: experimental prokaryote growth, uptake, and secretion rates. **Media**: citation-backed, model-ready growth media mapped to BiGG exchanges. All at [github.com/omidard](https://github.com/omidard)

## **SELECTED PUBLICATIONS AND INTELLECTUAL PROPERTY**

- **Annotating the pangenome reveals the diversity in the genetic basis for metabolic enzymes.** *Science Advances* 12(27) (2026). doi:10.1126/sciadv.aeb3363. First author.
- **BiGG 2026: a pangenome-scale metabolic-model generation pipeline and the next-generation release of the BiGG Models database.** *In preparation*. First author and pipeline architect.
- **Pangenome reconstruction of Lactobacillaceae metabolism predicts species-specific metabolic traits.** *mSystems* 9(7):e00156-24 (2024). doi:10.1128/msystems.00156-24. First author.
- **Rare metabolic gene essentiality is a determinant of microniche adaptation in *E. coli*.** *Under review*. Open code: [github.com/omidard/EcopanGEM](https://github.com/omidard/EcopanGEM). First author.
- **Phylon decomposition of the *E. coli* pangenome by non-negative matrix factorization.** *Under review*. Co-author.
- **Reconstruction and validation of a genome-scale metabolic model of *L. lactis* NCDO 2118; in-silico succinate and GABA overproduction.** *Biochemical Engineering Journal* 170:107967 (2021). doi:10.1016/j.bej.2021.107967. First author.
- **Lactobacillaceae-related patent. Main inventor; Notice of Invention filed, DTU.**
- Full list of peer-reviewed papers, preprints, and PhD thesis: ORCID 0000-0002-0064-3802

## **EDUCATION**

- **Ph.D., Systems Biology, Technical University of Denmark (DTU)** (2021-2024). Thesis: *Pangenome-scale metabolic reconstruction of the Lactobacillaceae family and Escherichia coli*. Supervisor: Prof. Bernhard O. Palsson.
- **M.Sc., Microbial Biotechnology, University of Tehran** (GPA 18.14 / 20, 2015-2018). Thesis: genome-scale model of *Lactococcus lactis* NCDO 2118; model-guided media optimization to increase GABA production.
- **B.Sc., Cellular and Molecular Biology / Biotechnology, Islamic Azad University, Arak** (GPA 17.53 / 20, 2010-2014). Trained across general biology, molecular biology, and biotechnology before moving into computational microbiology and systems biology, so the computational work rests on the underlying biology rather than on data alone.

## **LANGUAGES AND HONORS**

English (IELTS Academic 7.5) | Danish (basic, in progress) | Farsi (native) | Kurdish (native). Ranked 3rd nationally in the Iranian national entrance examination. Ranked 1st among M.Sc. students in Microbial Biotechnology, University of Tehran (2018).

## **REFERENCES**

Available on request. Prof. Bernhard O. Palsson (UCSD); Prof. Lars K. Nielsen (University of Queensland); Dr. Patrick V. Phaneuf (DTU).